

Propositional Logic

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CS 22 2021

May 17, 2021

Overview

Propositions from Propositions (3.1)

NOT, AND, and OR (3.1.1)

IMPLIES (3.1.2)

If and Only If (3.1.3)

Truth tables

Truth tables help convey how to interpret logical statements.

P	$\text{NOT}(P)$
F	T
T	F

P	Q	$P \text{ AND } Q$
F	F	F
F	T	F
T	F	F
T	T	T

Like a multiplication table, but flattened (so easier to apply to more inputs.)

OR

P	Q	$P \text{ OR } Q$
F	F	F
F	T	T
T	F	T
T	T	T

In English, often rendered as “ P or Q or both”. Inclusive or.

P	Q	$P \text{ XOR } Q$
F	F	F
F	T	T
T	F	T
T	T	F

Exclusive or, too.

OR examples

“Someday I want to visit London (X) or Paris (Y).”

X	Y	sentence
F	F	F
F	T	T
T	F	T
T	T	T

X OR Y

“With seatbelts, it’s click it (X) or ticket (Y).”

X	Y	sentence
F	F	F
F	T	T
T	F	T
T	T	F

X XOR Y

More OR Examples

“The recipe calls for milk chocolate (X) or peanut (Y) M&Ms.”

X	Y	sentence
F	F	F
F	T	F
T	F	T
T	T	T

X

“You may select the chicken (X) or the beef (Y).”

X	Y	sentence
F	F	T
F	T	T
T	F	T
T	T	F

NOT(X AND Y) aka X NAND Y

Implication

Perhaps counterintuitive.

X	Y	$X \text{ IMPLIES } Y$
F	F	T
F	T	T
T	F	F
T	T	T

Claim: If $\sqrt{2}$ is a rational multiple of π , then $(2x)^2$ is even for all integers x .

True or false? True. $X \text{ IMPLIES } Y$ is guaranteed to be true if Y is true.

Definition: In “ $X \text{ IMPLIES } Y$ ”, X is called the *hypothesis* and Y the *conclusion*.

$X \text{ IMPLIES } Y$ is the same as $Y \text{ OR NOT}(X)$.

Truth table for iff

X	Y	X IFF Y
F	F	T
F	T	F
T	F	F
T	T	T

Example: A number is divisible by three *if and only if* the sum of its digits is divisible by three.

Definitions have this form. Can also be written $\text{NOT}(X \text{ XOR } Y)$.
Or $(X \text{ AND } Y) \text{ OR } (\text{NOT}(X) \text{ AND } \text{NOT}(Y))$.